

# Engineering Mechanics Lab 2029

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Course code 2029, Engineering Mechanics Lab, Semester 2, Engineering Science, No Credit, 3 periods per week, 45 periods per semester.

Programme: Architecture, Automobile, Civil and allied, Mechanical and allied, Wood and Paper Technology, Environmental Engineering.

Objectives: hands-on experience in engineering mechanics and experiments with force systems.

Prerequisites: Mathematics I and Applied Physics I.

Outcomes: CO1 identify force systems; CO2 determine unknown forces; CO3 infer centre of gravity and mass moment of inertia; CO4 determine strain and coefficient of friction.

CO-PO mapping: CO1 to CO4 strongly mapped to PO4.

Source limitation: detailed official model practical paper pattern is not specified in supplied syllabus..

# Roadmap and Modules

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Module 1 CO1 15 hours: apparatus identification, resultant of concurrent force system using force table, graphical resultant of concurrent and parallel force systems, Lami theorem using Jib crane.

Draw force table, vector polygon, space diagram, force triangle.

Calculate resultant, equilibrant, direction and percentage error.

Module 2 CO2 6 hours: support reactions of simply supported beam, graphical support reactions using link polygon.

Draw beam, loads, reactions, load line and link polygon.

Module 3 CO3 9 hours: centroid of laminae, mass moment of inertia of flywheel and shaft, law of moments using bell crank lever.

Module 4 CO4 11 hours: tensile strain, coefficient of friction on horizontal plane and inclined plane, open ended experiments and Lab Exam II.

Common mistakes: wrong scale, missing units, parallax, zero error and confused resultant/equilibrant..

# Formula Bank and Solved Examples

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Resultant of two forces:  $R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$ .

Direction:  $\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$ .

Lami theorem:  $\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$ .

Beam equilibrium:  $\sum V = 0$ ,  $\sum M = 0$ .

Moment =  $M = Fx$ .

Centroid:  $\bar{x} = \frac{\sum Ax}{\sum A}$ ,  $\bar{y} = \frac{\sum Ay}{\sum A}$ .

Rotational energy =  $\frac{1}{2} I \omega^2$ .

Strain =  $\frac{\Delta L}{L}$ .

Horizontal friction:  $\mu = \frac{F}{N}$ .

Inclined plane:  $\mu = \tan \theta$ .

Solved beam: span 4 m, loads 200 N at 1 m and 300 N at 3 m.

$R_B \times 4 = 200 \times 1 + 300 \times 3 = 1100$ ,  $R_B = 275 \text{ N}$ ,  $R_A = 225 \text{ N}$ .

Friction:  $F = 18 \text{ N}$ ,  $N = 50 \text{ N}$ ,  $\mu = 0.36$ .

Strain: extension 0.24 mm, length 200 mm, strain = 0.0012.

Bell crank:  $80 \times 0.15 = F_2 \times 0.24$ ,  $F_2 = 50 \text{ N}$ .

# Practice, Model Paper and References

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Practice questions: define resultant, equilibrant, moment, centroid, strain and coefficient of friction.

Explain force table experiment, link polygon method, plumb line method and bell crank lever.

Descriptive questions: verify Lami theorem, determine beam reactions, determine coefficient of friction on inclined plane.

Model practical paper: Q1 identify five apparatus and uses, 5 marks.

Q2 viva definitions, 5 marks.

Q3 determine beam reactions with two point loads OR determine concurrent resultant graphically and verify using force table, 20 marks.

Q5 observation table and percentage error, 10 marks.

Q6 precautions, sources of error, industrial application and conclusion, 10 marks.

References: Khurmi Applied Mechanics, Bansal Engineering Mechanics, Ramamrutham Engineering Mechanics, Bedi Strength of Materials, Meriam and Kraige Statics, NPTEL and Virtual Labs..